

AE 16: Model Selection

Tips Data

Driver: _____, Reporter: _____, Gopher: _____

! Important

- Open [RStudio](#) and create a subfolder in your AE folder called “AE-16”.
- Go to the [Canvas](#) and locate your AE-16 assignment to get started.
- Upload the `ae-16.qmd` and `tip-data.csv` files into the folder you just created. The `.qmd` and PDF responses are due in Canvas. You can check the due date on the Canvas assignment.

Packages + data

```
library(tidyverse)
library(broom)
library(yardstick)
library(ggformula)
library(patchwork)
library(knitr)
library(kableExtra)

tips <- read_csv("tip-data.csv") |>
  drop_na(Party) |>
  select(-`Tip Percentage`, -`W/Tip`)
```

What factors are associated with the amount customers tip at a restaurant? To answer this question, we will use data collected in 2011 by a student at St. Olaf who worked at a local restaurant.¹

¹Dahlquist, Samantha, and Jin Dong. 2011. “The Effects of Credit Cards on Tipping.” Project for Statistics 212-Statistics for the Sciences, St. Olaf College.

The variables we'll focus on for this analysis are

- **Tip:** amount of the tip
- **Party:** Number of people in the party
- **Meal:** Time of day (Lunch, Dinner, Late Night)
- **Age:** Age category of person paying the bill (Yadult, Middle, SenCit)
- **Day:** Day of the week
- **Payment:** the type of payment used
- **GiftCard:** whether a gift card was used
- **Comps:** Whether any food was comped
- **Alcohol:** Whether any alcohol was ordered
- **Bday:** whether it was someones birthday
- **Bill:** the size of the bill

Analysis goal

The goals of this activity are to:

- Compare subsets of predictors using nested F-Tests
- Use best subset, forward, and backward selection to choose models

Exercise 0

Fit two models:

1. `tip_fit_1`: predict `Tip` from `Party`, `Age`, and `Meal`
2. `tip_fit_2`: predict `Tip` from `Party`, `Age`, `Meal`, and `Day`.

Exercise 1

Of the two models you just fit, which is nested inside the other?

Exercise 2

Fit a third model called `tip_fit_3` so that `tip_fit_1` is nested inside `tip_fit_3`. Try to choose a model that you think will do a better job than `tip_fit_1` of modeling the data. Feel free to include interaction or polynomial terms. Have your reporter write the model on the board.

Exercise 3

Change `eval: FALSE` to `eval: TRUE`. Alter the code below so that you're comparing `tip_fit_3` to `tip_fit_1`.

```
anova(tip_fit_1, tip_fit_2) |> # Enter reduced model first
  tidy() |>
  kable()
```

Write down the hypotheses, test statistic, and p-value of your test. Interpret the output in the context of the problem. Have your reporter write the p-value on the board.

Exercise 4

Change `eval: FALSE` to `eval: TRUE`. First, look at the documentation for `regsubsets` and determine what `nvmax` does. Alter the code below so that the largest model `regsubsets` will consider has 20 (wow!) terms in it. Find the “best” model according to the following criteria, and have your reporter write the model on a white board. Note: `arrange` will sort the rows so that the optimal model will appear at the top or bottom.

- Group 1: Find the best model according to R^2_{adj} .
- Group 2: Find the best model according to C_p/AIC .
- Group 3: Find the best model according to BIC .
- Group 4: Find the best model according to R^2 . (Don't write it up but think about if it's what we would expect)

```
library(leaps)

best_subsets <- regsubsets(Tip ~ ., data = tips)

best_subsets |>
  tidy() |>
  arrange(desc(____)) |> # Delete desc if you want to sort in decreasing order
  kable()

coef(best_subsets, ___) # Enter the number of predictors in the best model
```

Exercise 5

The `regsubsets` function has an argument called `method`. This argument tells `regsubsets` whether to use **exhaustive** (i.e. best subsets) selection, **forward stepwise**, or **backward stepwise** selection. Re-purpose your code above to find the best model according to the following criteria and have your reporter write down the terms that are included in your model (you do not need to write out the coefficients):

- Group 1: Forward stepwise selection using C_p .
- Group 2: Backward stepwise selection using C_p .
- Group 3: Forward stepwise selection using BIC .
- Group 4: Backward stepwise selection using BIC .

To submit the AE

! Important

- Render the document to produce the PDF with all of your work from today's class.
- Upload your QMD and PDF files to the Canvas assignment.